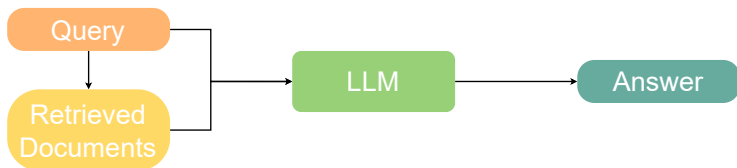
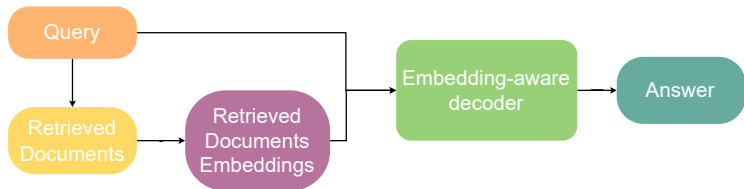


PISCO, OSCAR: compression methods for efficient retrieval-augmented generation (RAG)

Maxime Louis, Hervé Dejean, Thibault Formal, Stéphane Clinchant



- ▶ **Quadratic inference cost:** Transformer cost $\propto (\text{context length})^2$
- ▶ **Limited context window:** LLMs cap at $\sim 10\text{K} - 100\text{K}$ tokens
- ▶ **Accuracy vs. efficiency:** More docs improve QA but increase latency



- ▶ Documents are compressed into compact vector representations
- ▶ The LLM works on these representations
- ▶ Compression can be computed offline and stored (PISCO) or online (OSCAR)

Hard Compression[†]

Text → shorter text

- Pruning
- Extractive summarization

Hard Compression

[Chirkova et al., 2025, Xu et al., 2023]

Pros:

- ▶ LLM-agnostic (works with any model)

Cons:

- ▶ Limited ratio (only 2–5×)

Soft Compression

(Text → embeddings, up to 256×)

- Embeddings
- K/V caches

Soft Compression

Pros:

- ▶ High compression rate (up to 256×)

Cons:

- ▶ Model-specific (requires embedding-aware LLM)

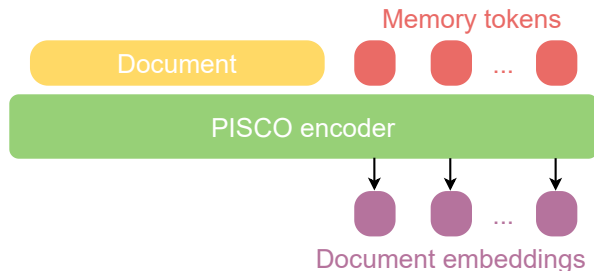
=> **The focus of this presentation**

Method	Backbone	Compression	ASQA	HotpotQA	NQ	TriviaQA	POPQA	Average
Decoders with no compression								
Mistral-7B	Mistral-7B	–	0.74	0.51	0.69	0.92	0.70	0.71
Llama-3.1-8B	Llama-3.1-8B	–	0.71	0.50	0.65	0.90	0.68	0.69
Compression Models								
xRAG [Cheng et al., 2024]	Mistral-7B	x128	0.34	0.27	0.32	0.77	0.33	0.40 (-0.31)
AutoCompressor [Chevalier et al., 2023]	Llama-2-8b	x4	0.57	0.31	0.35	0.70	0.24	0.43 (-0.26)
ICAE [Ge et al., 2023]	Llama-2-8b	x4	0.47	0.29	0.42	0.78	0.43	0.48 (-0.21)
DODO [Qin et al., 2024]	Llama-2-8b	x10	0.52	0.38	0.48	0.82	0.47	0.53 (-0.16)
COCOM [Rau et al., 2024]	Mistral-7B	x16	0.63	0.46	0.58	0.89	0.48	0.61 (-0.10)

- All current methods suffer from strong performance degradation,
- Additionally, all existing methods require medium-scale pretraining.

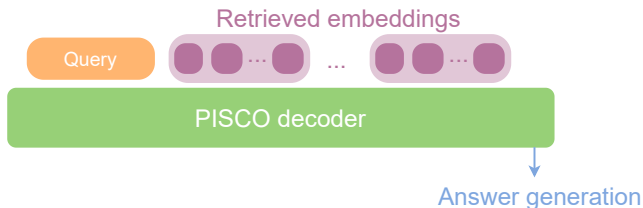
PISCO: offline compression for RAG

07/2025



- Embeddings are the last hidden states at each learnable memory token appended to an input document.
- After training, **compression done offline and stored on the document collection** for maximum inference efficiency.
- As in [Rafailov et al., 2024, Chevalier et al., 2023]:

$$\mathbf{c}_i = (c_i^s)_{s=1\dots l} = \mathcal{F}_{\theta_c}(d_i, m_1, \dots, m_l)$$



- Generation is conditioned on the document embeddings $\mathbf{c}_1, \dots, \mathbf{c}_k$:

$$a_i \sim \mathcal{F}_{\theta_d}(\cdot \mid q, \mathbf{c}_1, \dots, \mathbf{c}_k, \mathbf{a}_{<i}),$$

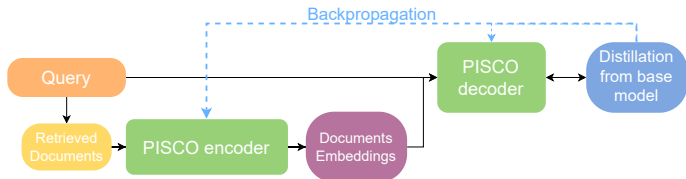
- Effective context length is reduced by the compression rate.

- ▶ **Working on compressed representations should not change the generations** => use distillation. With teacher:=(standard RAG model), student:=(compressed RAG model).
- ▶ xRAG [Cheng et al., 2024] and CEPE [Yen et al., 2024] use "token-level" distillation:

$$D_{\text{KL}}\left(p_{\phi}(X_{\text{answer}} \mid X_{\text{context}}, \cdot) \parallel p_{\phi}(X_{\text{answer}} \mid W(E_{\text{context}}), \cdot)\right)$$

where X_{answer} is the **ground-truth** answer,

- ▶ **In contrast we use sequence-level distillation:** cross-entropy loss on teacher-generated labels.



- **Sequence-level distillation from a teacher LLM:** generate teacher labels from text documents:

$$a_i \sim \mathcal{T}(\cdot \mid d_1, \dots, d_k, q, \mathbf{a}_{<i}).$$

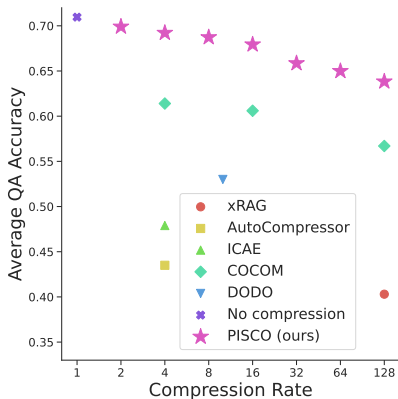
- **End-to-end training with:**

$$\mathbf{c}_i = (c_i^s)_{s=1, \dots, l} = \mathcal{F}_{\theta_c}(d_i, m_1, \dots, m_l)$$

$$\mathcal{L}(\theta_c, \theta_d) = - \sum_{i=1}^r \log \mathcal{F}_{\theta_d}(a_i \mid q, \mathbf{c}_1, \dots, \mathbf{c}_k, \mathbf{a}_{<i})$$

- ▶ **Backbones:** Mistral-7B, Llama-3.1-8B, SOLAR-10.7B, with LoRA adapters,
- ▶ Strong retrieval (Splade-v3) and reranking (Deberta-v3) pipeline,
- ▶ **Train set:** A single epoch on 453 k QA pairs (Wikipedia-KILT + MSMarco + Squad), $k = 5$ top documents,
- ▶ **Compression rate:** $\times 16$; (also ablate $\times 2 \rightarrow 128$),
- ▶ The teacher LLM is Mistral-7B, with greedy decoding.

PISCO main results (accuracy)

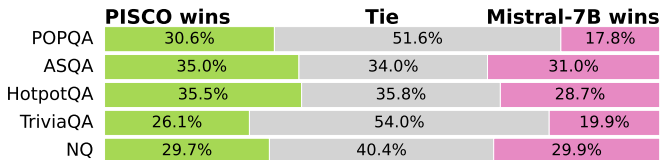
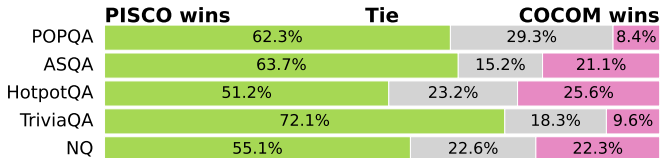


Model	GFlops	Time
Mistral-7B	11232	0.08s
Pisco-Mistral	2803	0.02s

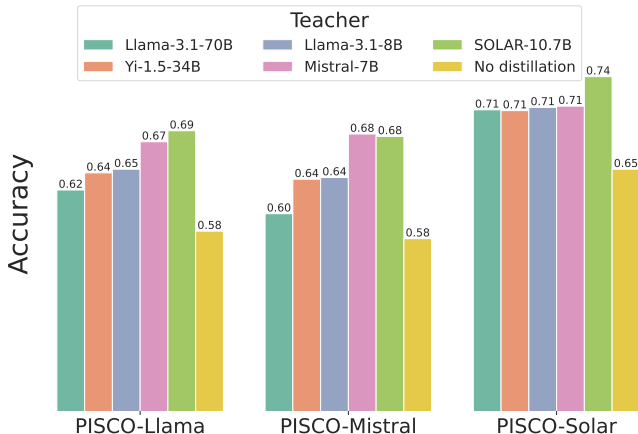
Efficiency gains.

- PISCO has very limited losses on the in-domain benchmarks, here measured using the **accuracy** metric,
- Models available on Huggingface: [naver/pisco-mistral](https://huggingface.co/naver/pisco-mistral).

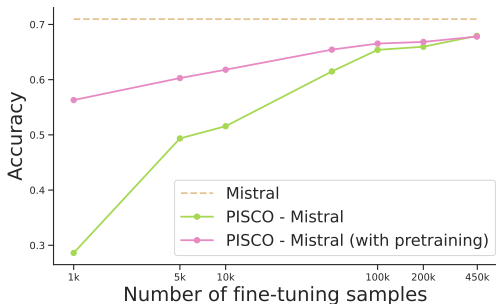
Pairwise evaluations with gpt-4o



- ▶ PISCO-Mistral outperforms COCOM [Rau et al., 2024],
- ▶ It matches the performance of its Mistral-7B backbone.

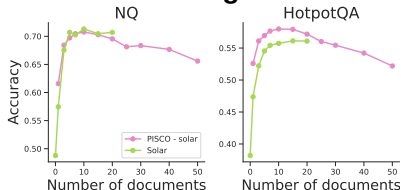


- Distillation is key to obtain strong results,
- Best teacher is not always the largest model.



- ▶ Additional pretraining tasks (AE, text continuation, keyword prompts, multi-doc...) on large datasets **do not lead to downstream QA improvements**,
- ▶ Pretraining aids performance only with <100k fine-tuning samples; negligible benefit at 450k samples,
- ▶ Pretraining is only necessary when no distillation is used.

Document Scaling



- ▶ Trained on $k = 5$ retrieved docs: but maintain performance when tested at higher k

- ▶ **Summarization** (CNN/DailyMail): PISCO-Mistral $\simeq$ Mistral.

Out-of-Domain QA

Model	Bio-QA	Covid	ParaphraseRC	RobustQA Avg
Llama	0.26	0.17	0.48	0.24
PISCO-Llama	0.24	0.12	0.38	0.27
Mistral	0.27	0.13	0.49	0.26
PISCO-Mistral	0.26	0.11	0.38	0.26
Solar	0.28	0.14	0.50	0.26
PISCO-Solar	0.29	0.10	0.47	0.27

- ▶ Out-of-domain datasets (Bio-QA, Covid, etc. with different document collections from kilt)
- ▶ PISCO generalizes on par with its backbones

- ▶ PISCO achieves high compression ($\times 16$) with minimal accuracy loss,
- ▶ Outperforms or matches existing soft compression baselines,
- ▶ Generalizes across QA, summarization, and out-of-domain tasks,
- ▶ Requires no large-scale pretraining, only distillation.
- ▶ Collection of embeddings to be stored: hardware/infrastructure burden => OSCAR

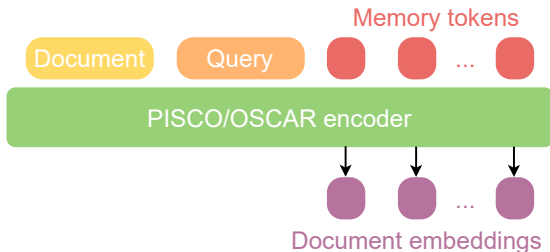
OSCAR: online query-dependent compression for RAG

07/2025

- ▶ To work online, compression needs to be fast,
- ▶ Replace the generator-size compressor with either:
 - A smaller LLM e.g. Llama-1B or ModernBert,
 - Just the first few-layers of the generator backbone.

Online compression can use the query

- If compressing online, we can use the user query



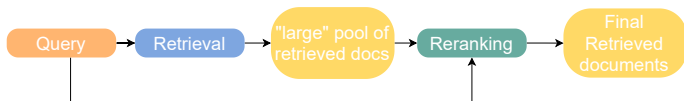
- Provides gains for large compression rates:

Model	ASQA	HotpotQA	NQ	TriviaQA	POPQA	BIOASQ
PISCO (x128)	0.81	0.57	0.75	0.89	0.51	0.77
OSCAR-llama (x128)	0.81	0.61 ↑0.04	0.79 ↑0.04	0.90 ↑0.01	0.63 ↑0.12	0.77

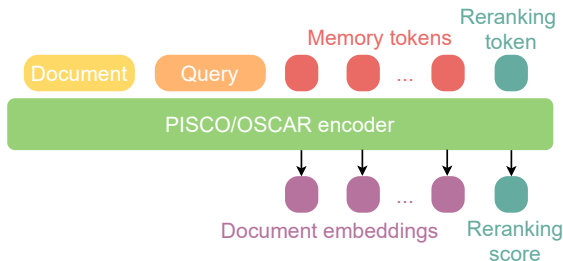
Reranking comes for free with OSCAR

[Chirkova et al., 2025]

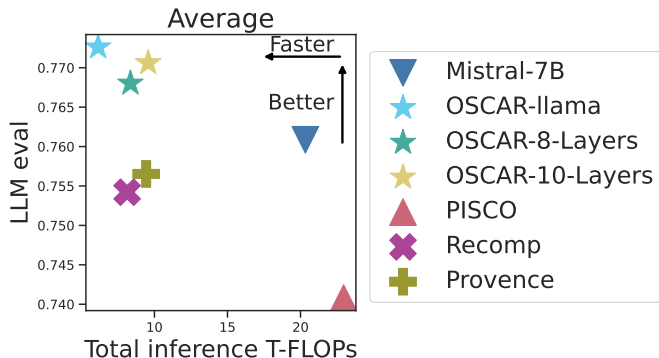
- Strong RAG pipelines use a 2-stage retrieval:



- OSCAR compression is a cross-encoding operation: we can add a re-ranking head to rerank for free:



- ▶ Mostly identical to PISCO,
- ▶ Reranking is learned by distillation (mse) from DeBERTa-v3 scores,
- ▶ OSCAR-Llama models need medium-scale pretraining to align the representations of encoder/decoder,
- ▶ Tested on other backbones: Mistral-7B, Mistral-24B and Qwen2.5-7B.



- ▶ Reranking recovers (almost perfectly) teacher performance,
- ▶ Efficiency gains: 2x-5x,
- ▶ Little to no accuracy loss on all tested backbones,
- ▶ Best compressor is in fact ModernBert-large (slightly faster, as good as llama).

Dataset	Trained decoder	Frozen decoder
ASQA	0.72	0.65
HotpotQA	0.48	0.42
NQ	0.65	0.72
TriviaQA	0.90	0.87
POPQA	0.66	0.51

- ▶ All attempts at fine-tuning only the encoder model (to use generic LLM for generations) fall short,
- ▶ Context length limitations,
- ▶ Compression rate flexibility,
- ▶ Multilingual support.

Thank you for your attention



Cheng, X., Wang, X., Zhang, X., Ge, T., Chen, S.-Q., Wei, F., Zhang, H., and Zhao, D. (2024).

xrag: Extreme context compression for retrieval-augmented generation with one token.

arXiv preprint arXiv:2405.13792.



Chevalier, A., Wettig, A., Ajith, A., and Chen, D. (2023).

Adapting language models to compress contexts.

arXiv preprint arXiv:2305.14788.



Chirkova, N., Formal, T., Nikoulina, V., and Clinchant, S. (2025).

Provence: efficient and robust context pruning for retrieval-augmented generation.

arXiv preprint arXiv:2501.16214.



Ge, T., Hu, J., Wang, X., Chen, S.-Q., and Wei, F. (2023).

In-context autoencoder for context compression in a large language model.

arXiv preprint arXiv:2307.06945.



Qin, G., Rosset, C., Chau, E., Rao, N., and Van_Durme, B. (2024).

Dodo: Dynamic contextual compression for decoder-only lms.

Proceedings of the 62nd Annual Meeting of the Association for Computational



Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. (2024).

Direct preference optimization: Your language model is secretly a reward model.

Advances in Neural Information Processing Systems, 36.



Rau, D., Wang, S., Dejean, H., and Clinchant, S. (2024).

NAVER LABS

Context embeddings for efficient answer generation in rag.

arXiv preprint arXiv:2407.09252.



Xu, F., Shi, W., and Choi, E. (2023).

Recomp: Improving retrieval-augmented lms with compression and selective augmentation.

arXiv preprint arXiv:2310.04408.



Yen, H., Gao, T., and Chen, D. (2024).

Long-context language modeling with parallel context encoding.

arXiv preprint arXiv:2402.16617.